Towards Trustworthy Breast Cancer Prognosis: Integrating Uncertainty-Aware Survival Prediction and Patient-Centred Communication

How can an uncertainty-aware Individual Survival Distribution framework be adapted for breast cancer prognosis to provide calibrated, patient-specific survival predictions with model-related uncertainty, and communicate these outputs through a patient-centred dashboard?

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This dissertation investigates the development of an uncertainty-aware framework for personalised breast cancer prognosis. Using the METABRIC dataset, an MTLR-based Individual Survival Distribution approach was combined with X-CAL-style calibration-aware optimisation and ensemble disagreement to produce calibrated patient-specific survival predictions alongside an indicator of model-related uncertainty. These outputs were integrated into a patient-centred dashboard designed to present survival estimates, uncertainty and explanatory information in an accessible form. Computational evaluation demonstrated that calibration could be improved while largely preserving discrimination, while ensemble disagreement was associated with individual prediction error despite not improving overall predictive performance. The resulting system demonstrates the feasibility of integrating prediction reliability, uncertainty awareness and patient-centred presentation within a single breast cancer prognostic research prototype.

1 INTRODUCTION

1.1 Background

Breast cancer prognosis is influenced by heterogeneous clinical, pathological and molecular characteristics, making accurate estimations of individual outcomes important for treatment planning, patient management and communication. Established prognostic systems such as PREDICT demonstrate how patient and tumour characteristics can be combined to estimate survival following breast cancer surgery (Wishart et al., 2010). Advances in molecular profiling have also provided increasingly detailed information about tumour biology, supporting a shift towards more personalised approaches to prognosis (Nicolini et al., 2018).

Machine-learning methods provide additional flexibility for modelling complex relationships between prognostic factors and survival outcomes. However, increasing predictive sophistication does not necessarily produce more trustworthy clinical information. Personalised survival estimates remain probabilistic, and their usefulness depends not only on predictive performance but also on whether probabilities are appropriately calibrated, whether uncertainty is represented, and how predictions are communicated to intended users. As a result, developing clinically meaningful prognostic systems requires consideration of both prediction quality and the interpretation of model outputs, providing the motivation for an uncertainty-aware survival framework.

1.2 Problem Statement

Increasingly personalised survival predictions do not necessarily provide sufficiently reliable or interpretable prognostic information. Survival models are frequently assessed through discrimination, yet strong patient ranking does not ensure that predicted survival probabilities are appropriately calibrated. Haider et al. (2020) demonstrates that evaluation of individual survival distributions requires complementary measures of discrimination and calibration, while Goldstein et al. (2020) further shows that distributional calibration can be explicitly incorporated into survival-model optimisation.

Calibration alone also does not describe the uncertainty associated with an individual prediction. Variation arising from model selection and estimation can influence survival predictions, encouraging approaches that account for model uncertainty rather than relying exclusively on a single fitted model (Thamrin et al., 2013). Further, producing calibrated predictions and uncertainty estimates does not guarantee that these outputs can be appropriately interpreted when presented to users. The problem is therefore not solely one of improving predictive performance, but of integrating personalised survival prediction, calibration, model-related uncertainty and responsible communication within a coherent prognostic system.

1.3 Research Gap

Existing research provides established approaches for individual survival prediction, distributional calibration, model uncertainty, uncertainty visualisation and human-centred clinical AI. However, these concerns are commonly addressed as separate methodological or communication problems. For example, Haider et al. (2020) focuses on constructing and evaluating ISDs, while Goldstein et al. (2020) addresses explicit distributional calibration through X-CAL. Additionally, uncertainty visualisation and healthcare-interface research considers how complex predictive information can be presented and interpreted by intended users.

As a result, there remains limited research integrating patient-specific survival distributions, calibration-aware modelling, model-related predictive uncertainty and patient-centred communication within a unified breast cancer prognostic framework. This creates a research gap between methods concerned with the statistical reliability of personalised survival predictions and those concerned with how such predictions and their limitations are presented to users. This dissertation addresses that gap through the development and evaluation of an uncertainty-aware ISD framework applied to breast cancer prognosis and integrated within a patient-centred dashboard.

1.4 Research Aim

The aim of this dissertation is to adapt an uncertainty-aware Individual Survival Distribution framework for breast cancer prognosis, integrating calibrated patient-specific survival prediction, model-related uncertainty estimation and patient-centred communication within a decision-support dashboard.

1.5 Research Objectives

To achieve the research aim, the following objectives were established:

1. Investigate existing approaches to breast cancer survival prediction, uncertainty estimation, calibration and patient-centred communication.
2. Adapt an Individual Survival Distribution framework using MTLR for personalised breast cancer prognosis using the METABRIC dataset.
3. Incorporate calibration-aware optimisation to improve the reliability of patient-specific survival distributions.
4. Estimate and represent uncertainty using ensemble disagreement between independently trained survival models.
5. Design and implement a patient-centred dashboard presenting personalised survival predictions, ensemble disagreement and supporting explanatory information.
6. Evaluate the proposed framework using computational performance measures and examine how the resulting predictions and uncertainty estimates can be presented through a patient-centred dashboard.

1.6 Structure

The remainder of this dissertation is organised into six chapters. Chapter 2 reviews literature relating to survival analysis, breast cancer prognosis, calibration, uncertainty and patient-centred communication. Chapter 3 describes the research methodology, including the prediction framework, dataset, evaluation strategy and ethical considerations. Chapter 4 presents the implementation and integration of the computational framework and dashboard, including the principal design decisions and their justification. Chapter 5 explains the computational evaluation and resulting prognostic outputs, while Chapter 6 discusses these findings in relation to the existing literature and considers the limitations of the study. Finally, Chapter 7 summarises the research contributions, answers the research question and identifies directions for future work.

2 LITERATURE REVIEW

Literature was identified through searches of PubMed, IEEE Xplore, and Google Scholar, with sources selected according to their relevance to breast cancer prognosis, survival modelling, calibration, predictive uncertainty and patient-centred communication. Preference was given to peer-reviewed research introducing or evaluating methods directly relevant to the proposed framework, alongside established clinical and methodological literature. A key challenge was that these topics are largely addressed within separate research communities. As a result, the review synthesises evidence across survival analysis, machine learning, healthcare communication and human–computer interaction rather than treating prediction performance in isolation.

2.1 Survival Analysis

Survival analysis models the time until an event of interest, such as death, recurrence or metastasis. Unlike conventional classification, survival modelling accounts for both event timing and censoring, where the target event has not been observed. There are two key quantities used to characterise survival data, the survival function which

describes the probability of remaining event-free beyond a given time, and the hazard function which represents the instantaneous rate at which the event occurs among individuals who have survived up to the given time (Collett, 2023).

Traditional analysis approaches include the Kaplan–Meier estimator and the Cox proportional hazards model. Cox regression became widely used in clinical research because it supports multivariable analysis without requiring a fixed parametric baseline hazard, while producing interpretable hazard ratios. However, its proportional hazards assumption can be restrictive. Bellera et al. (2010) showed in breast cancer that prognostic effects such as tumour grade and hormone-receptor status may vary over time, meaning that a conventional Cox model can obscure clinically relevant temporal patterns.

The increasing availability of high-dimensional clinical, genomic and multimodal data has therefore encouraged the use of machine-learning and deep-learning survival models capable of representing nonlinear relationships and complex interactions (Wiegrebe et al., 2024). Despite this, greater complexity does not automatically imply better clinical use. Comparative work has shown that deep models can achieve strong predictive performance, while simpler and more interpretable approaches may remain competitive (Wang et al., 2024).

Therefore, modern survival analysis is better viewed as an expanding methodological toolkit rather than a replacement of classical statistics. This progression has shifted attention from population-level risk estimation towards personalised prognosis, providing the foundation for contemporary cancer survival prediction.

2.2 Cancer Survival Prediction

Cancer survival prediction uses survival modelling to estimate individual prognosis, supporting treatment planning, patient counselling and clinical decision-making. Breast cancer provides a particularly relevant setting because prognosis is influenced by heterogeneous, clinicopathological, and molecular characteristics. Historically, prognostic assessment relied primarily on factors such as tumour size, lymph-node involvement and histological characteristics, with biomarkers subsequently contributing both prognostic and treatment-predictive information (Nicolini et al., 2018). Established tools such as PREDICT demonstrate how these factors can be combined computationally to estimate survival following breast cancer surgery (Wishart et al., 2010).

Advances in molecular profiling have expanded this approach beyond conventional clinical variables. Contemporary cancer datasets increasingly incorporate genomic, transcriptomic and other molecular measurements, enabling models to characterise biological heterogeneity that clinical variables alone may not capture. Tran et al. (2025) identifies a corresponding shift from predominantly clinical prognostic models towards approaches integrating molecular and clinical information, with outputs including metastasis-free and progression-free survival.

Machine-learning and deep-learning approaches have consequently become increasingly prominent in breast cancer prognosis, offering greater flexibility for modelling complex relationships between patient characteristics. Kalafi et al. (2019), for example, evaluated neural networks alongside random forests, decision trees and support-vector machines for breast cancer survival prediction. However, methodological sophistication alone does not guarantee clinically useful predictions. Differences in feature selection, missing-data handling, dataset characteristics and validation procedures complicate comparison and may limit generalisability across populations.

Cancer survival prediction has therefore progressed towards increasingly personalised and biologically informed modelling. However, producing an individual prediction does not establish how reliable that prediction is, hence the need for uncertainty-aware survival modelling.

2.3 Uncertainty-Aware Survival Prediction

Predictive performance alone is insufficient for trustworthy healthcare AI because models may produce an apparently precise prediction despite substantial uncertainty. This is particularly important in clinical applications, where incomplete data, distribution shifts and model limitations can affect prediction reliability. Lechuga López et al. (2025) argues that uncertainty quantification (UQ) is important for improving the robustness, reliability and interpretability of these systems.

UQ commonly distinguishes between aleatoric and epistemic uncertainty. Aleatoric uncertainty shows inherent variability or noise within the data-generating process and is generally unable to be eliminated through additional training data. Epistemic uncertainty instead shows uncertainty in the model's knowledge, from factors such

as limited training data or poor representation, and therefore can be reduced with additional data or improved modelling (Lechuga López et al., 2025).

Within survival prediction, probabilistic and Bayesian approaches have been proposed to supplement personalised predictions with uncertainty estimates. For example, Loya et al. (2019) extends MTLR using Bayesian and Bayesian-neural formulations, enabling patient-specific survival predictions to be accompanied by uncertainty estimates. Model averaging and ensemble approaches provide an alternative by examining variation across multiple fitted models, representing uncertainty arising from model selection or parameter estimation (Thamrin et al., 2013).

However, uncertainty alone does not establish whether predicted survival probabilities correspond appropriately to observed outcomes.

2.4 Calibration

Discrimination and calibration capture different aspects of predictive performance. Discrimination describes a model's ability to correctly rank patients by risk, commonly assessed in survival analysis using the concordance index. However, discrimination does not ensure that predicted probabilities accurately reflect observed outcomes. Calibration instead considers agreement between predicted and observed event probabilities, making it particularly important where prognostic estimates may inform clinical decisions or patient communication (Van Calster et al., 2016).

Calibration is more complex in survival analysis as predictions vary over time and outcomes may be censored. Assessing calibration at a single time point therefore provides limited information about the reliability of the complete predicted survival distribution. Haider et al. (2020) addressed this through D-Calibration, which evaluates calibration across the full distribution of predicted event times rather than at one fixed horizon. Goldstein et al. (2020) extended this to X-CAL, a differentiable approximation to D-Calibration that allows calibration to be explicitly optimised during model training. Their experiments demonstrated that D-Calibration could be improved without necessarily producing a substantial reduction in concordance or likelihood.

However, calibration should not be considered independently of other aspects of prediction quality. Avati et al. (2020) argues that useful survival distributions should also be sharp, concentrating probability around plausible event times while maintaining calibration. As a result, survival-model evaluation requires consideration of complementary properties rather than optimisation of discrimination alone. While calibration establishes whether probabilistic predictions are statistically reliable at the population level, personalised prognosis additionally requires an appropriate representation of survival over time.

2.5 Individual Survival Distribution

Conventional survival models often summarise prognosis using risk scores or survival probabilities at predetermined time points. Haider et al. (2020) highlights that survival probabilities may change substantially between selected time points, potentially producing different interpretations depending on the horizon chosen. They instead advocate Individual Survival Distribution (ISD) models, which generate a complete patient-specific survival curve representing the probability of surviving beyond each future time point.

ISDs provide several advantages for personalised prognosis. A complete survival distribution enables survival probabilities to be obtained at different clinically relevant horizons without requiring separate models, while quantities such as median survival can also be derived. Importantly, the resulting predictions can be evaluated beyond just discrimination. Haider et al. (2020) propose evaluating ISDs using complementary measures including concordance, calibration, Brier score and D-Calibration, with the latter assessing calibration across the complete predicted survival distribution. This is significant because good discrimination does not necessarily imply that the probabilities represented by an individual survival curve are reliable.

However, richer predictions also create additional requirements. ISDs require more comprehensive evaluation than single risk scores, and different metrics capture distinct properties of prediction quality; for example, D-Calibration and Integrated Brier Score are not interchangeable measures (Haider et al., 2020). Additionally, producing a complete patient-specific distribution does not ensure that the information can be interpreted appropriately by patients or clinicians.

2.6 Communicating Medical Uncertainty

Uncertainty is inherent in clinical decision-making, particularly when population-level evidence is used to estimate an individual patient's prognosis. Communicating this uncertainty is key for patient-centred and shared decision-making tools because presenting probabilistic predictions without their limitations can create a misleading impression of certainty. Simpkin and Armstrong (2019) argue that transparent discussion of uncertainty is important for informed decision-making and the patient–clinician relationship, while acknowledging that uncertainty remains infrequently communicated in practice.

However, uncertainty communication presents its own challenges. Patients differ in comprehension, preferences and tolerance of uncertainty, while the way probabilistic information is framed can influence its interpretation and subsequent decisions. Risk information presented using different numerical formats can produce different perceptions despite describing the same underlying message (Naik, 2012). Additionally, communicating uncertainty is not universally beneficial, as it may increase worry, perceived vulnerability or decision dissatisfaction for some patients, demonstrating that transparency alone does not guarantee effective communication (Simpkin and Armstrong, 2019).

Effective communication therefore requires balancing transparency with comprehensibility rather than either concealing uncertainty or overwhelming patients with technical information. Importantly, there remains no universally optimal format for presenting uncertainty, and whether verbal, numerical, graphical or combined approaches are most effective remains an active research question.

2.7 Uncertainty Visualisation

Visualisation provides a means of communicating probabilistic information that may be difficult to interpret numerically, with approaches including probability distributions, intervals, density representations and visual encodings of variability (Kamal et al., 2021). Within prognostic systems, graphical representations can combine multiple risk factors and present their implications in a form intended to support communication and decision-making. Murray et al. (2018), for example, demonstrated this through graphical presentation of prognostic probabilities following traumatic brain injury.

However, visualising uncertainty does not automatically make it understandable. Different representations can affect interpretation, while excessive visual complexity may introduce additional burden or misinterpretation. Kinkeldey et al. (2014) found that comparatively little uncertainty-visualisation research had empirically evaluated proposed techniques with users, emphasising the importance of evaluating visualisations according to the tasks they are intended to support. The appropriate representation is therefore dependent not only on the underlying uncertainty but also on the intended audience and context.

For personalised survival prediction, this creates a need to present changing survival probabilities and uncertainty over time without obscuring the central prognosis. Therefore, effective visualisation should be considered as part of the wider interface rather than as an isolated graphical problem.

2.8 Patient-Centered Dashboard Design

Predictive models do not communicate directly with users, their outputs are interpreted through an interface. Dashboard design should therefore be considered part of the whole predictive system rather than a cosmetic final stage. Healthcare dashboards combine information to support monitoring, interpretation and decision-making, but their design requirements depend on the intended users and clinical context. In predictive healthcare, Szymanski et al. (2024) found that explanations can be too complex for non-expert users and emphasised user-centred design to improve their understanding and usefulness.

Patient-centred dashboards consequently require a balance between information richness and comprehensibility. Clear information hierarchy, simplified representations and contextual explanations can reduce unnecessary complexity while retaining important information. Szymanski et al. (2024), for example, found that six of eight healthcare professionals preferred simplified feature-importance representations over more detailed alternatives.

Accessibility must also be considered as a design requirement. Srinivasan et al. (2023) demonstrates how meaningful descriptions, explicit interaction controls and structured navigation can improve dashboard accessibility for screen-reader users.

For a prognostic dashboard, these principles imply that survival predictions and uncertainty should be presented with clear hierarchy, supporting explanation and sufficient detail without overwhelming the main prognosis. However, usability and accessibility alone do not establish whether users will appropriately interpret or rely upon an AI-generated prediction.

2.9 Explainable AI & Trust in Clinical Decision Support

Trust in clinical AI cannot be inferred from predictive performance alone. Explainable AI (XAI) attempts to make model predictions more understandable by exposing information about how outputs were produced, while broader transparency also encompasses uncertainty, limitations and intended use. This is particularly relevant to clinical decision support, where users must be able to critically assess predictions rather than accept them solely because they originate from a computational model. However, existing XAI research contains substantial gaps in real-world evaluation of explanation fidelity, usability and clinician trust (Abbas et al., 2025).

Importantly, the objective should not be to maximise trust, but to support appropriate reliance. Sivaraman et al. (2023) found that clinicians interacted with AI recommendations in different ways, including ignoring, negotiating with and relying upon them, while explanatory visualisations improved perceived usefulness but did not necessarily change agreement with AI recommendations. Explanations can also introduce risks. Panigutti et al. (2023) notes that XAI may increase over-reliance on flawed clinical decision-support systems, demonstrating that greater explainability does not automatically produce safer decision-making.

Human-centred explanation design is therefore necessary to align information with the needs of intended users rather than technical developers alone. This distinction reinforces that explainability, calibration and uncertainty are complementary rather than interchangeable. Explanation concerns understanding a prediction, calibration concerns probabilistic reliability, and uncertainty concerns limitations in predictive confidence. A trustworthy prognostic system must therefore integrate these dimensions with accessible communication rather than treating any single property as sufficient.

2.10 Literature Synthesis

The literature demonstrates substantial progress across survival prediction, uncertainty quantification, calibration and the communication of clinical AI outputs. However, these areas are frequently addressed as separate problems. Survival modelling research has increasingly moved towards personalised predictions, with ISDs providing complete patient-specific survival trajectories rather than isolated risk estimates (Haider et al., 2020). Calibration research has simultaneously demonstrated that discrimination alone is insufficient, and that survival probabilities must also correspond appropriately with observed outcomes, while methods such as X-CAL enable calibration to be incorporated into model optimisation (Goldstein et al., 2020). Avati et al. (2020) further demonstrates that calibration should be considered alongside sharpness, since calibrated but excessively broad predictions may provide limited individual prognostic information.

Calibration does not, however, represent the uncertainty associated with an individual prediction. Model averaging provides an established mechanism for accounting for model uncertainty within survival analysis (Thamrin et al., 2013), while wider healthcare research emphasises the importance of representing uncertainty when predictions support consequential decisions (Lechuga López et al., 2025). Consequently, a survival model may require consideration of discrimination, calibration and predictive uncertainty as complementary dimensions rather than interchangeable measures of reliability.

Other literature addresses how this information should be communicated. Research on medical uncertainty demonstrates that transparent communication can support informed decision-making, but may also increase worry or dissatisfaction depending on its presentation (Simpkin and Armstrong, 2019). Similarly, uncertainty visualisation and dashboard research demonstrates that presenting additional information does not inherently improve understanding, as effectiveness depends on representation, audience and interface design. Explainability can create tension, as

explanations may support interpretation while also encouraging inappropriate reliance on AI recommendations (Panigutti et al., 2023).

The resulting research gap is therefore primarily integrative rather than algorithmic. Existing research provides methods for individual survival prediction, calibration, uncertainty estimation and patient-centred communication, but there is limited work combining these components within a unified breast cancer prognostic system. This dissertation addresses this gap by adapting an established ISD framework to breast cancer, incorporating calibration-aware modelling and model-related uncertainty estimation, and communicating the resulting personalised survival predictions through a patient-centred dashboard.

3 METHODOLOGY

3.1 Research Design

This study adopted an applied computational research design to investigate the adaptation and integration of established methods for uncertainty-aware breast cancer survival prediction. Rather than proposing a novel survival-learning algorithm, the research combined an existing Individual Survival Distribution (ISD) framework with calibration-aware optimisation and model-based uncertainty estimation. The resulting predictions were then integrated into a patient-centred dashboard, allowing both the computational and communication components of the proposed system to be evaluated.

The research followed a staged experimental strategy comprising dataset preparation, survival-model development, calibration-aware training, uncertainty estimation, dashboard integration and evaluation. This approach reflects the interdisciplinary nature of the research question, combining survival analysis and machine learning with principles from uncertainty communication and human-computer interaction. Computational evaluation considered complementary aspects of prediction quality, including discrimination, calibration and uncertainty, while the dashboard was assessed against literature-informed accessibility, uncertainty-communication and human-centred design principles.

3.2 Prediction Framework

The prediction framework followed the Individual Survival Distribution (ISD) paradigm described and evaluated by Haider et al. (2020), with Multi-Task Logistic Regression (MTLR) used as the underlying survival model. Rather than generating a single risk score or fixed-horizon probability, MTLR was used to produce a complete patient-specific survival distribution across ordered time intervals. This aligns with the ISD framework, which advocates complete personalised survival trajectories where prognosis is required across multiple time points.

The MTLR-based framework was adapted to breast cancer using clinical and molecular predictors from the METABRIC dataset, with survival time and censoring status defining the time-to-event outcome. The resulting patient-specific survival curve provides the foundation for both quantitative evaluation and dashboard visualisation. This distributional representation was particularly appropriate because it enabled prognosis and associated uncertainty to be considered across time rather than restricting communication to a single interval.

The MTLR foundation was then extended through calibration-aware optimisation and ensemble modelling. Calibration was informed by X-CAL, which uses a differentiable approximation to D-Calibration to explicitly incorporate distributional calibration within model training (Goldstein et al., 2020). Ensemble modelling was then used to estimate predictive uncertainty through variation between the independently trained models. The framework therefore combined MTLR-based individual survival prediction, calibration-aware optimisation and uncertainty estimation as complementary components of the proposed prognostic system.

3.3 Dataset

The system used data from the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) cohort. METABRIC was selected because it combines breast cancer-specific clinical and molecular characteristics with survival

outcomes, making it suitable for investigating personalised prognosis. The dataset has also been used extensively in computational breast cancer research, including multimodal survival-prediction studies.

Following data auditing, 1,981 of 2,509 patient records contained the required overall-survival information and were eligible for modelling. Of these, 1,144 represented observed deaths and 837 were censored/living observations. Overall survival was defined using OS_MONTHS and OS_STATUS. The final extended model contained 13 predictors: age at diagnosis, cellularity, positive lymph nodes, ER status, HER2 status, PR status, grade, tumour size, tumour stage, inferred menopausal state, histological subtype, integrative cluster and non-synonymous tumour mutational burden. [See Figure A1]

Eligible patients were divided into 60% training, 20% validation and 20% held-out test sets, separated by event status. Numerical predictors were median-imputed and standardised, while categorical predictors were mode-imputed and one-hot encoded, with preprocessing fitted using the training data to reduce information leakage.

However, METABRIC represents a retrospective historical cohort, meaning that treatment practices, missing data and population representation may limit generalisability to contemporary breast cancer populations.

3.4 Evaluation

The prediction framework was evaluated using complementary measures of discrimination, overall prediction error, calibration and uncertainty. Multiple measures were required because no individual metric captures all properties of an Individual Survival Distribution. Haider et al. (2020), for example, demonstrates that D-Calibration and Integrated Brier Score assess distinct properties and that a model may perform well according to one while performing poorly according to the other.

Discriminative performance was assessed using Harrell's concordance index (C-index), which measures the extent to which predicted prognostic ordering agrees with observed survival outcomes (Harrell et al., 1996). Higher values indicate stronger discrimination, although concordance does not determine whether the predicted survival probabilities themselves are accurate. Overall probabilistic prediction error was therefore assessed using the Integrated Brier Score (IBS), which evaluates prediction error across time while considering the distributional nature of survival predictions. IBS is an established component of ISD evaluation (Haider et al., 2020).

Distributional calibration was evaluated using D-Calibration, which tests calibration across the complete predicted survival distribution rather than at an isolated time point (Haider et al., 2020). This was particularly relevant because the calibration-aware training procedure was informed by X-CAL, which seeks to improve D-Calibration while balancing calibration against predictive performance (Goldstein et al., 2020).

Predictive uncertainty was evaluated using between-model disagreement within the MTLR ensemble. Rather than interpreting disagreement as a formal confidence interval, the analysis examined whether patients receiving higher ensemble disagreement also tended to receive higher prediction error. This tested whether the uncertainty signal contained useful information about model stability at the individual-patient level.

3.5 Ethics

This study used an existing anonymised breast cancer dataset and involved no recruitment of patients, collection of new clinical data or clinical intervention. Analysis was restricted to de-identified METABRIC data required for survival modelling, with no attempt to identify participants. The project therefore involved secondary analysis of existing research data rather than direct human-participant research.

Ethical considerations nevertheless extend beyond data privacy. Prognostic predictions concern potentially sensitive information and may be misinterpreted if presented as deterministic outcomes. The developed system was therefore treated as a research prototype rather than an actual clinical system. Survival estimates were presented as probabilistic predictions, with uncertainty and limitations made explicit within the interface. This is supported by wider concerns surrounding clinical AI that systems should assist rather than replace professional judgement and that responsibility for clinical decisions should remain clearly defined (Harwich & Laycock, 2018).

Potential bias and generalisability were also considered when interpreting model performance. METABRIC represents a retrospective cohort, meaning that cohort composition, historical treatment practices and available variables may influence both predictions and estimated uncertainty. As a result, performance on the study population was not interpreted as evidence of validity across contemporary breast cancer populations or other clinical settings.

The system has not undergone prospective clinical validation and is not intended for clinical diagnosis, treatment recommendation or independent patient decision-making.

3.6 Chapter Summary

This chapter has described the methodology used to adapt and evaluate an uncertainty-aware ISD framework for breast cancer prognosis, including the research design, MTLR-based prediction framework, METABRIC dataset, evaluation strategy and ethical considerations. The following chapter describes how these methodological components were integrated within the computational artefact and justifies the principal implementation and interface-design decisions.

4 IMPLEMENTATION

The computational artefact was implemented as a full-stack web application comprising a Python machine-learning backend and a React/TypeScript frontend. PyTorch supported the MTLR, calibration-aware and ensemble modelling pipeline, with FastAPI providing the REST API connecting the predictive service to the frontend. React Router supported interface navigation and Recharts was used for the interactive survival visualisation. Automated testing was conducted across both components: pytest covered preprocessing, survival modelling, calibration, ensemble-disagreement processing, prognosis generation and API behaviour, while Vitest and React Testing Library tested frontend rendering, validation, interaction workflows, state management and selected accessibility behaviours. TypeScript compilation, production builds and linting provided additional static and build-time verification.

4.1 Architecture Description

The computational artefact was implemented as an end-to-end pipeline connecting breast cancer survival modelling, calibration, ensemble disagreement estimation and prognostic visualisation. Following preprocessing of the METABRIC data, patient characteristics were provided to an MTLR model that generated an Individual Survival Distribution (ISD), representing predicted survival probability across the modelled time horizon. Calibration-aware optimisation was incorporated during model training to improve the reliability of these distributional predictions.

Multiple independently trained MTLR models were combined as an ensemble to provide an indicator of model-related predictive uncertainty. The central prediction was calculated as the mean of ensemble-member survival probabilities at each prediction time, while ensemble disagreement was quantified using the population standard deviation of these probabilities. An overall patient-level disagreement score was calculated by averaging the time-specific standard deviations across the evaluated survival trajectory. The resulting mean survival curve was restricted to remain within valid probability bounds and to be non-increasing over time.

These outputs were then sent to the dashboard as the communication layer of the artefact. The ensemble mean formed the central patient-specific survival curve, while model disagreement was visualised using a band representing the mean prediction ± one standard deviation, bounded between zero and one. This was explicitly treated as an ensemble disagreement band rather than a statistical confidence interval. The complete architecture therefore connected METABRIC preprocessing → MTLR-based ISD prediction → calibration-aware optimisation → ensemble disagreement estimation → patient-specific prognostic visualisation.

4.2 Design Decisions

Several design decisions were made to balance predictive capability, calibration and interpretability. MTLR was selected because it directly supports distributional survival prediction while remaining compatible with the ISD paradigm adopted in this study. Survival time was divided using a target of 20 quantile-based bins derived exclusively from uncensored training-set event times. Rather than using equally spaced temporal intervals, event-time quantiles were selected to distribute observed deaths more evenly across the MTLR intervals, with a final open-ended interval representing survival beyond the final boundary. Alternative configurations of 10, 20 and 30 bins were considered during model development.

Calibration-aware optimisation was incorporated using an X-CAL-style objective, introducing a trade-off between improving distributional calibration and potentially affecting discrimination or overall predictive performance. Early stopping was applied according to validation performance to limit unnecessary training and reduce overfitting, with the best validation objective occurring at epoch 65. [See Figure A2]

An ensemble was selected as a practical means of representing model-related uncertainty without requiring a fully Bayesian survival model. Variation between independently trained MTLR models was represented through ensemble disagreement, while their mean provided the central prediction. Standard deviation was selected as a time-dependent measure of disagreement but was deliberately not presented as a confidence interval because it does not provide formally calibrated statistical coverage. This distinction informed both the terminology and visual representation used within the artefact.

Interface design prioritised communicating the primary prognosis without hiding uncertainty. The patient-specific survival curve therefore formed the principal visualisation, accompanied by the ensemble disagreement band and selected time-horizon probabilities. Progressive disclosure separated the primary prognosis from more detailed explanations, reducing initial information density while retaining access to interpretation guidance and model limitations.

Plain-language explanations and accessibility considerations were also incorporated to reduce the likelihood of probabilistic outputs being interpreted as deterministic forecasts. These design decisions were intended to support theoretical patient-centred interpretation, although their effectiveness was not empirically established because no patient or clinician evaluation was conducted.

4.3 Design Justification

The artefact was designed to operationalise the requirements identified within the literature rather than treating survival prediction, calibration, uncertainty and communication as independent problems. The use of MTLR within an ISD framework reflects Haider et al.'s (2020) argument for patient-specific survival distributions that provide probabilities across time rather than only risk scores or fixed-horizon estimates. Calibration-aware training draws upon Goldstein et al. (2020), whose X-CAL approach demonstrates that distributional calibration can be incorporated directly into model optimisation and balanced against predictive performance.

The ensemble component addresses the limitation of uncertainty arising from reliance on a single fitted model. Thamrin et al. (2013) demonstrate that conditioning inference on one selected survival model can ignore model uncertainty and motivate combining information across models. Although the present implementation uses independently trained MTLR models rather than Bayesian model averaging, the underlying principle motivates examining variation between plausible fitted models. Ensemble disagreement was therefore incorporated as an indicator of model-related uncertainty, rather than being interpreted as a complete measure of prognostic uncertainty.

Finally, the dashboard operationalised the communication principles identified in the literature review by combining the survival trajectory, ensemble disagreement and explanatory information within a single interface. The objective was not only to expose additional model outputs, but to distinguish the central prediction from its limitations and avoid presenting probabilistic prognosis as certainty. As a result, the artefact integrates personalised distributional prediction, calibration, model disagreement and patient-centred communication within a single system. However, because no patient or clinician evaluation was conducted, the design can only be claimed to implement literature-informed communication principles, not to demonstrate improved comprehension or decision-making.

5 RESULTS & EVALUATION

5.1 Model Evaluation

Model development and configuration were conducted using the training and validation partitions, with the held-out test set reserved for final evaluation. Successive configurations were compared on the validation partition to assess the effects of extending the predictor set, calibration-aware optimisation and ensemble modelling. The clinical Cox baseline achieved a C-index of 0.659, compared with 0.672 for the extended MTLR, 0.671 following X-CAL-style

regularisation and 0.667 for the ensemble [see Figure A3]. The extended MTLR achieved an IBS of 0.1620 and adjusted D-Calibration χ^2 of 9.189, while X-CAL regularisation produced values of 0.1617 and 5.407 respectively. The ensemble achieved an IBS of 0.1638 and adjusted D-Calibration χ^2 of 11.898. X-CAL therefore improved distributional calibration with minimal change in discrimination and a small improvement in prediction error.

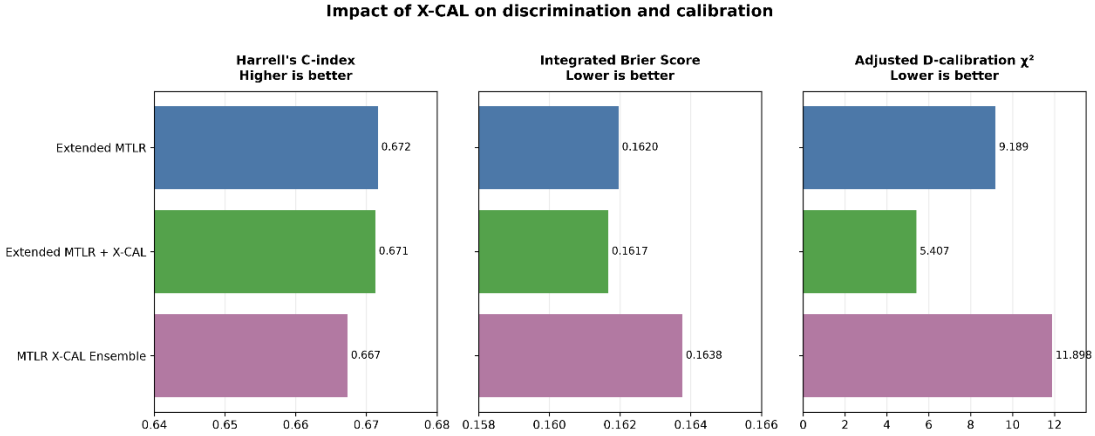


Figure 1 – Impact of X-CAL on discrimination and calibration

On the validation partition, ensemble disagreement was strongly positively associated with absolute prediction error at 12 months (Spearman's $\rho = 0.925$, $p < 0.001$), 36 months ($\rho = 0.834$, $p < 0.001$) and 60 months ($\rho = 0.688$, $p < 0.001$). The association weakened with increasing prediction horizon but remained strong across all three time points [see Figure A4].

Following model selection, the X-CAL-regularised MTLR and X-CAL ensemble were evaluated on the previously unseen held-out test partition (n = 397), without using test performance to modify the models or preprocessing. The X-CAL MTLR achieved a C-index of 0.666, IBS of 0.1725 and adjusted D-Calibration χ^2 of 3.292 ($p = 0.952$), while the ensemble achieved 0.662, 0.1737 and 4.745 ($p = 0.856$), respectively. Both models showed no evidence of distributional miscalibration at the 0.05 significance level. Ensemble disagreement also remained strongly associated with test-set prediction error at 12 months ($\rho = 0.905$, $p < 0.001$), 36 months ($\rho = 0.807$, $p < 0.001$) and 60 months ($\rho = 0.690$, $p < 0.001$).

Overall, held-out performance was consistent with the principal validation findings. Discrimination decreased only slightly, although IBS indicated moderately greater prediction error on unseen patients. The ensemble did not outperform the single X-CAL model on aggregate predictive measures, but its disagreement signal retained a strong relationship with individual prediction error. These findings support the use of complementary evaluation measures rather than selecting a model according to discrimination alone.

5.2 Dashboard and Prognostic Output

The final outputs were integrated into a patient-centred prognostic dashboard, putting the underlying survival distributions into an interactive presentation. For an individual patient, the dashboard presents the predicted survival trajectory together with survival probabilities at selected 12-, 36- and 60-month horizons. The central survival curve represents the mean prediction across ensemble members, while the surrounding disagreement band displays variation between those models. An overall disagreement indicator additionally summarises average ensemble disagreement across the evaluated trajectory.

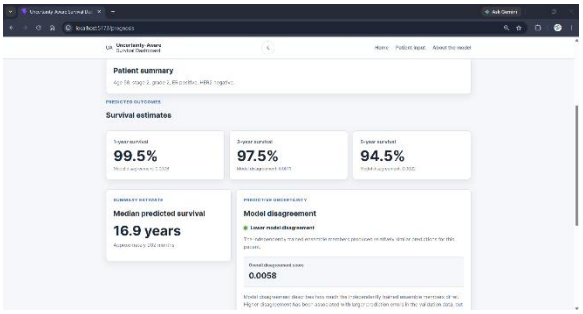


Figure 2- Final patient-centred prognosis dashboard



Figure 3- Individual survival distribution and ensemble disagreement visualisation

The interface distinguishes the predicted prognosis from the uncertainty associated with the underlying models. Ensemble disagreement is explicitly described as model disagreement rather than a statistical confidence interval, avoiding an unsupported interpretation of the band as having defined probabilistic coverage. Supporting explanations and limitations are provided through progressive disclosure, so that additional information can be accessed without overwhelming the primary prediction with technical details. The dashboard also clarifies that survival estimates are probabilistic and that the prototype is not intended to replace clinical judgement.

Accessibility was incorporated at both structural and communication levels. Pages use semantic HTML, logical heading hierarchies, labelled regions and explicitly associated form labels, while validation errors are programmatically linked to affected controls and announced to assistive technologies. Keyboard interaction is supported through visible focus states and skip navigation, with reduced-motion and responsive-layout behaviour additionally accommodating different interaction and viewing requirements. The survival visualisation is provided with an accessible textual description, while its principal information is not dependent on the graph or colour alone: survival probabilities are repeated numerically and model disagreement is communicated through textual explanations, numerical values and graphical representation. This redundancy was particularly important for uncertainty communication, where interpretation should not depend solely upon perceiving the disagreement band.

Accessibility also informed the organisation and complexity of the information presented. The patient-input workflow groups related variables and provides explanatory text for less familiar predictors, followed by a structured review of entered values before prognosis generation. Within the output, progressive disclosure prioritises the principal prognosis while retaining access to the complete survival trajectory and more detailed uncertainty information. Plain-language explanations accompany technical outputs, including the explicit qualification that ensemble disagreement is neither a statistical confidence interval nor a representation of all sources of clinical uncertainty.

The completed artefact therefore demonstrates the technical integration of patient-specific ISDs, selected prognostic horizons and model-disagreement information within a single interface. However, no patient or clinician usability study was conducted, therefore, these results demonstrate implementation of the intended communication approach rather than evidence that the dashboard improves comprehension, usability or clinical decision-making. [See Figure A5]

5.3 Critical Discussion of Results

The results demonstrate that model quality depended on the evaluation criterion considered. The extended MTLR achieved the strongest discrimination (C-index = 0.672), while X-CAL-style regularisation produced a similar C-index of 0.671 alongside a lower IBS and substantially improved adjusted D-Calibration χ^2, from 9.189 to 5.407. This suggests that improved distributional calibration was achieved with minimal loss of discrimination, reinforcing the importance of evaluating survival distributions using complementary metrics rather than selecting a model according to concordance alone.

The ensemble produced a different trade-off. The ensemble did not improve aggregate predictive performance, recording a lower C-index and poorer IBS and D-Calibration than the single X-CAL-regularised model. However, ensemble disagreement was strongly associated with absolute prediction error at 12, 36 and 60 months, with the association weakening at longer horizons. Ensemble disagreement therefore provided a potentially informative indicator of model instability even though the ensemble did not improve the aggregate evaluation metrics.

These findings establish that the implemented framework can generate patient-specific survival distributions, improve their distributional calibration through calibration-aware optimisation, and derive a model-disagreement signal related to prediction error. They do not, however, establish clinical validity, generalisability beyond METABRIC, or whether the dashboard improves patient or clinician understanding.

6 DISCUSSION

6.1 Interpretation of Findings

The results demonstrate that an MTLR-based ISD framework can be adapted to the METABRIC breast cancer setting to generate complete patient-specific survival distributions. However, the comparison between model configurations shows that predictive quality depends on more than discrimination. Although the extended MTLR achieved the strongest concordance, incorporating X-CAL-style regularisation substantially improved calibration with almost no reduction in discrimination and a small improvement in IBS. Selecting the model according to C-index alone would therefore have obscured improvements in the reliability of its predicted survival distributions.

The ensemble produced a different outcome. Combining independently trained models did not improve aggregate discrimination, calibration or prediction error compared with the X-CAL-regularised model. Nevertheless, greater ensemble disagreement was strongly associated with greater prediction error, particularly at shorter prediction horizons. This suggests that disagreement contains useful information about model instability at the individual-patient level, despite not improving overall predictive performance. It should not, however, be interpreted as a complete measure of prognostic uncertainty or as a calibrated prediction interval, since it only captures variation between ensemble members.

The dashboard further demonstrates that patient-specific survival trajectories, fixed-horizon probabilities and ensemble disagreement can be integrated within a single interface. Collectively, the findings support the technical feasibility of combining personalised survival prediction, calibration-aware modelling and uncertainty within the proposed framework. However, technical integration does not demonstrate clinical usability. Without patient or clinician evaluation, no conclusions can be drawn regarding whether the dashboard improves comprehension, usability, trust or decision-making.

6.2 Comparison with Literature

The findings broadly support Haider et al. (2020), who argues that ISDs provide a richer basis for personalised prognosis and require evaluation across complementary dimensions rather than discrimination alone. The present results reinforce this position as they show that the model with the highest C-index was not the model with the strongest distributional calibration. This distinction was particularly clear following calibration-aware optimisation, where the improvement in D-Calibration occurred with minimal change in discrimination. This is consistent with Goldstein et al. (2020), who demonstrate through X-CAL that distributional calibration can be explicitly optimised without sacrificing substantial predictive performance.

The results also support the broader argument that survival-prediction quality cannot be represented by a single metric. Avati et al. (2020) emphasises that useful survival distributions must balance complementary predictive properties, while the present findings demonstrate differences between discrimination, probabilistic prediction error and calibration. Thamrin et al. (2013) encourages accounting for uncertainty arising from reliance on a single fitted survival model. Although the ensemble did not improve aggregate predictive performance in this study, disagreement between independently trained models was strongly associated with individual prediction error. This suggests that

variation between models may provide useful information even where model averaging alone does not produce complete predictions.

Finally, incorporating this disagreement into the dashboard reflects the uncertainty-communication literature's emphasis on avoiding unsupported certainty in clinical information (Simpkin and Armstrong, 2019). However, uncertainty visualisation and XAI research also cautions that exposing additional information does not guarantee appropriate interpretation and may increase complexity or inappropriate reliance (Kinkeldey et al., 2014; Panigutti et al., 2023). The present artefact therefore extends the technical survival-modelling pipeline into a literature-informed communication interface but cannot establish whether this representation is effective without evaluation involving its intended users.

6.3 Limitations

Several limitations constrain the interpretation and generalisability of these findings. First, development and evaluation were conducted using a single retrospective METABRIC cohort without external or prospective validation. Historical treatment practices, cohort composition and demographic representation may therefore differ from contemporary clinical populations, while missing-data handling and preprocessing introduce additional assumptions. Additionally, performance within METABRIC should not be interpreted as evidence that equivalent performance would be achieved across other breast cancer populations or healthcare settings. The scope of model and hyperparameter exploration was also necessarily limited, therefore the configuration cannot be assumed to represent the optimal predictive architecture.

Second, the ensemble disagreement measure only captures variation between independently trained models and therefore represents a limited component of model-related uncertainty rather than total predictive uncertainty. The displayed disagreement band similarly has no calibrated statistical coverage and should not be interpreted as a confidence or prediction interval. Additionally, calibration-aware optimisation was informed by X-CAL rather than constituting an exact reproduction of the original method, meaning that comparisons with Goldstein et al. (2020) should be evaluated accordingly.

Finally, the dashboard remains a research prototype and was not evaluated with patients or clinicians. Although its design was informed by literature discussing uncertainty communication, visualisation, accessibility and patient-centred interfaces, the study provides no external evidence that these design choices improve comprehension, usability, trust or decision-making. The framework has likewise undergone no prospective clinical validation and should not be interpreted as suitable for clinical deployment or autonomous decision support. Future work should therefore prioritise external and prospective validation, more comprehensive uncertainty quantification, and formal evaluation of the interface with intended users before considering clinical application.

7 CONCLUSION

7.1 Summary

This dissertation investigated the integration of personalised survival prediction, calibration and model-related uncertainty within a breast cancer prognostic system. Using the METABRIC dataset, an MTLR-based implementation of the ISD paradigm was developed to generate complete patient-specific survival distributions. The framework incorporated X-CAL-style calibration-aware optimisation and an ensemble of independently trained models, with variation between ensemble members used as an indicator of model-related uncertainty. These outputs were then integrated into a literature-informed, patient-centred dashboard presenting survival trajectories, selected prognostic horizons and ensemble disagreement.

Evaluation demonstrated that different model configurations provided advantages across different dimensions of prediction quality. Calibration-aware optimisation substantially improved distributional calibration while largely preserving discrimination, demonstrating the importance of evaluating survival predictions beyond just concordance. Although the ensemble did not improve aggregate predictive performance, greater ensemble disagreement was strongly associated with greater individual prediction error. The completed artefact therefore

demonstrates the technical feasibility of integrating personalised survival distributions, calibration and model-disagreement information within a unified breast cancer prognostic framework, while its effectiveness for patient or clinical use remains unestablished without prospective and user-centred evaluation.

7.2 Contributions

The primary contribution of this dissertation is the integration of established survival-modelling and uncertainty-aware methods within a breast cancer prognostic framework, rather than the development of a novel survival-learning algorithm. Technically, the study demonstrates how an MTLR-based ISD model can be combined with calibration-aware optimisation and ensemble disagreement to produce patient-specific survival trajectories that consider discrimination, calibration and model-related uncertainty as complementary properties.

The study also provides a contribution by adapting this framework to the METABRIC breast cancer cohort and evaluating the resulting predictions using multiple survival-specific criteria. Finally, the computational outputs were translated into a literature-informed patient-centred dashboard that integrates survival trajectories, selected time-horizon probabilities, ensemble disagreement and explanatory information. This provides a prototype demonstrating how technically complex prognostic outputs can be incorporated within a single communication interface, although its effectiveness for patients or clinicians remains to be established empirically.

7.3 Research Question Reflection

This dissertation addressed the research question: "How can an uncertainty-aware Individual Survival Distribution framework be adapted for breast cancer prognosis to provide calibrated, patient-specific survival predictions with model-related uncertainty, and communicate these outputs through a patient-centred dashboard?"

The findings provide a qualified positive answer. An MTLR-based ISD framework was successfully adapted to breast cancer prognosis using METABRIC, producing complete patient-specific survival distributions. Incorporating X-CAL-style regularisation improved distributional calibration with minimal reduction in discrimination, while ensemble disagreement was strongly associated with individual prediction error despite the ensemble not improving aggregate predictive performance. Together, these findings demonstrate that calibrated personalised survival predictions and model-disagreement information can be integrated within a single computational framework.

The communication component of the research question can only be answered partially. The resulting survival predictions and ensemble disagreement were successfully incorporated into a literature-informed patient-centred dashboard, demonstrating the technical feasibility of presenting these outputs together. However, no patient or clinician evaluation was conducted and therefore, the study cannot establish whether the dashboard communicates prognostic uncertainty effectively, improves understanding, or supports decision-making in practice. The research question is therefore answered at the level of technical integration and computational evaluation, while the effectiveness of the proposed communication approach remains an area for empirical investigation.

The research objectives were consequently achieved to differing extents. The literature surrounding breast cancer prognosis, survival modelling, calibration, uncertainty and patient-centred communication was synthesised to inform the framework, and an MTLR-based ISD approach was successfully adapted to the METABRIC cohort. Calibration-aware optimisation was incorporated and evaluated, while ensemble disagreement was implemented as an indicator of model-related predictive uncertainty. These outputs were integrated within a patient-centred dashboard incorporating survival estimates, uncertainty information and explanatory content. Computational evaluation established the predictive and calibration characteristics of the framework. The final objective was achieved at the level of technical implementation and computational evaluation, however, the effectiveness of the communication approach was not empirically established because no patient or clinician evaluation was conducted. Therefore, the study demonstrates how the predictions and uncertainty information can be presented, but cannot establish the usability, comprehensibility or effectiveness of that presentation for its intended users.

7.4 Future Work

Future work should first evaluate the framework using external and contemporary breast cancer cohorts, followed by prospective validation to establish whether performance can be generalised beyond METABRIC. Further investigation

of model architectures, hyperparameters and formally calibrated uncertainty-quantification methods could also determine whether predictive performance and uncertainty estimation can be improved beyond its current implementation.

A particularly important next stage would involve formal evaluations with patients and clinicians to assess comprehension, usability, interpretation of uncertainty and appropriate reliance on the dashboard. Findings from these studies could support interface redesign based on observed user needs rather than solely literature-derived principles. The framework could subsequently be adapted and evaluated for other cancer types or time-to-event clinical problems. Longer-term research could also investigate integration with treatment-effect modelling to support treatment-specific prognostic comparisons. Ultimately, prospective clinical evaluation would be required before the system could be considered for real-world deployment.

Retrospectively, the development process highlighted the value of considering prediction quality, uncertainty and communication as requirements from the outset rather than sequential stages of model development. Earlier involvement of patients and clinicians would have enabled the dashboard to be effectively refined using empirical evidence rather than literature-derived principles alone. Future development would therefore benefit from a more iterative process in which modelling, uncertainty evaluation and user-centred design are evaluated concurrently.

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APPENDICES A

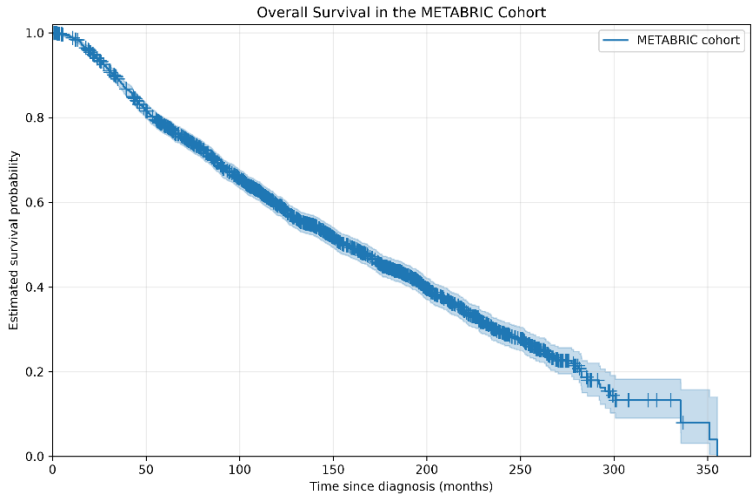


Figure A1- METABRIC Kaplan Meier

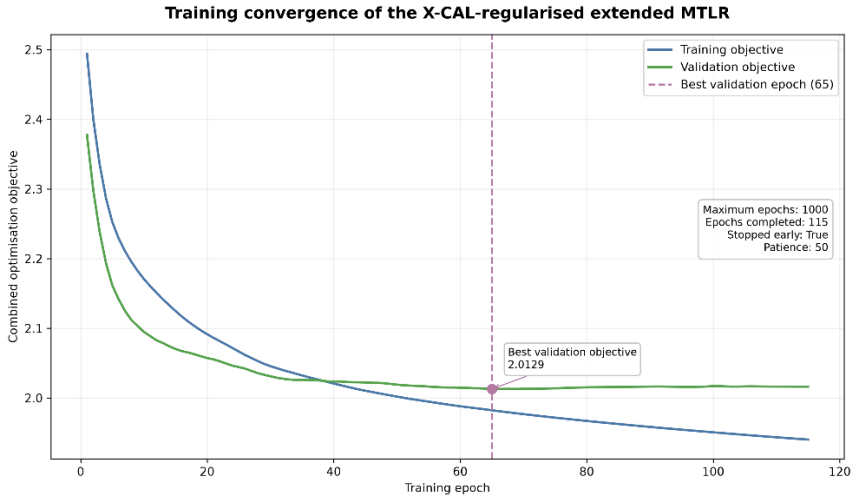


Figure A2- Training Convergence

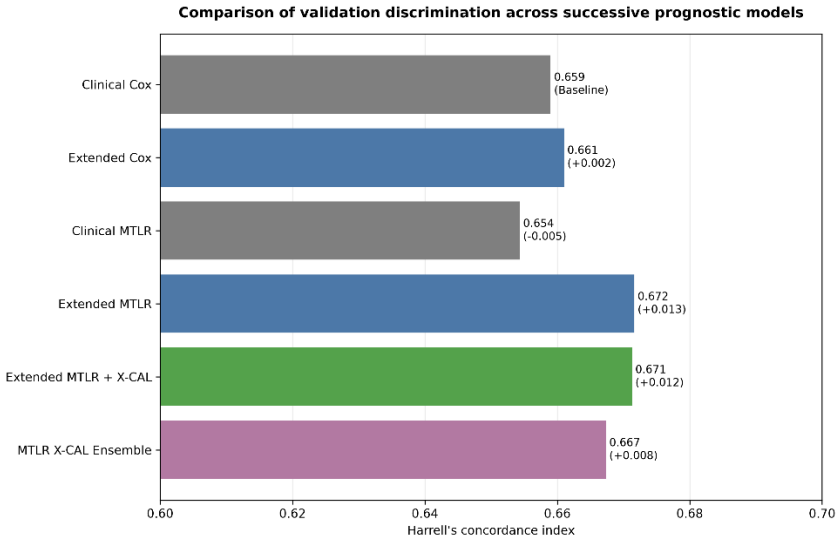


Figure A3- Model Comparison

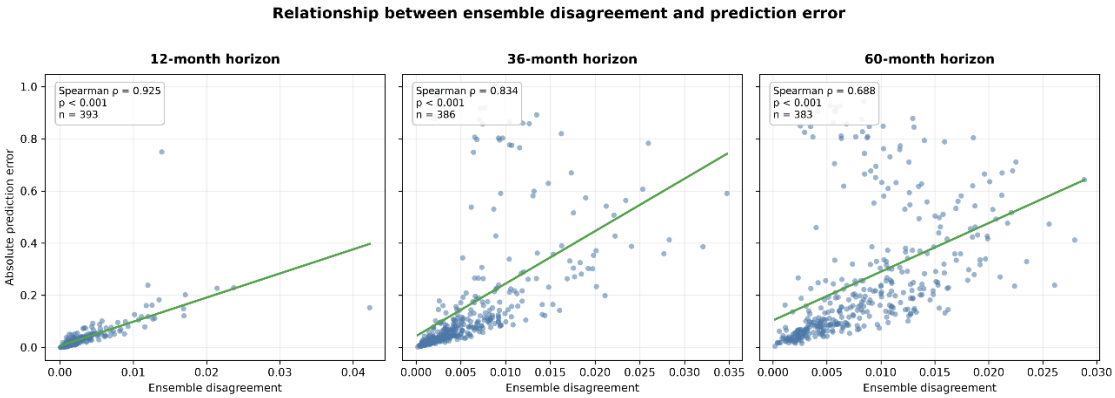


Figure A4- Uncertainty Error Relationship

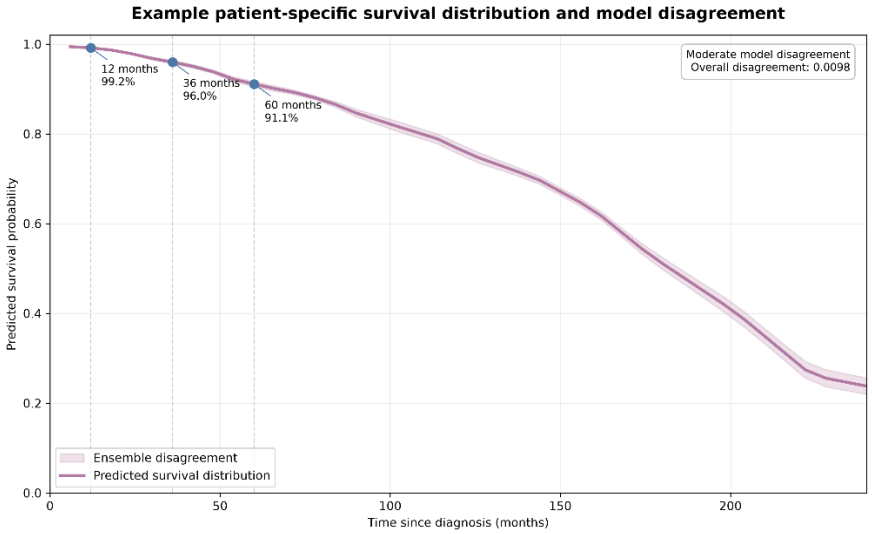


Figure A5- Example Patient Prognosis